Literature Review: Chatbot Using Recurrent Neural Networks (RNN)

Recent studies (2025–2026) with three earlier foundational studies (2022–2023)

Ref	Paper Title	Author(s)/Year	Journal Name	Month & Year of Publication	Methodology & Technology	Advantages	Limitations
1	AI Chatbot for Research Paper Development Using RNN	Karpaga Valli, S., Mekala Devi, S., SheikYousuf, T. & Manickavasagam, J. (2026)	Engineering Genesis (Shanlax), Vol. 1, Issue 2	Jan–Jun 2026 (issue period)	RNN/LSTM sequence-to-sequence chatbot with tokenization, normalization, context-management and response-generation modules; cloud-hosted; evaluated with accuracy, precision, recall, F1 and confusion matrix.	Keeps context across turns; supports keyword-based content generation, summarization and academic tone; reported response time under 2 s; more flexible than rule-based and retrieval-based tools.	Repetition in generated text; lower accuracy on highly domain-specific queries; dataset and numeric scores not clearly reported; transformer integration left as future work.
2	Adaptive Emotion-Aware Chatbot for Mental Health Diagnosis Using Recurrent Reinforcement Learning and Transformer Models	Dessai, S., Arora, S. & Joshi, G. (2026)	Frontiers in Artificial Intelligence, 9, 1769286	May 2026	LSTM/GRU recurrent policy trained with Proximal Policy Optimization (PPO) to build dynamic questionnaires, combined with transformer models for emotion awareness.	Adaptive questioning avoids redundant questions and reduces user burden; recurrent layers keep history-aware context through the questionnaire.	Hybrid design, so the RNN's standalone contribution is hard to isolate; heavier training pipeline (RL + transformer); limitations to be confirmed from the full text.
3	Multimodal Mental Health Chatbot	Harini, V. K., Suryanaryanan, A., Alex, L. & Bhavadharini, R. M. (2026)	Springer, Advances in AI and ML in Big Data Processing (AAIMB 2024), pp. 97–108	January 2026	Task-oriented, closed-domain, retrieval-based chatbot built on an LSTM-RNN; accepts voice and text input; compared with other embeddings and transformer architectures.	Reported 89.36% accuracy; multimodal (voice + text) input improves accessibility; focused domain gives relevant answers.	Retrieval-based, so it can only return stored responses; closed-domain scope; no generative flexibility.
4	Virtual Assistant Based on Recurrent Neural Networks: Scope and Coverage of Health Insurance	Paz-Medina, D. A., Rojas-Cavassa, D. E. & Carrera-Salas, E. A. (2025)	International Journal of Advanced Computer Science and Applications (IJACSA), Vol. 16, No. 10	October 2025	Simple RNN intent classifier (64-d embedding, 128 RNN units, softmax) trained on 109 FAQ pairs, combined with GPT-3.5 Turbo for response generation; web app (Angular, Django, AWS).	82% mean response accuracy on 50 test questions; user satisfaction 4.59/5; lightweight and fast to train; easy to adapt to other domains.	Simple RNN struggles with complex or extended dialogues; depends on an external OpenAI model; very small training set and only 32 test users; some answers unclear or too long.

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5	Seq2Seq Model-Based Chatbot with LSTM and Attention Mechanism for Enhanced User Interaction	Benaddi, L., Ouaddi, C., Souha, A., Jakimi, A., Rahouti, M., Aledhari, M., Oliveira, D. & Ouchao, B. (2025)	arXiv preprint (arXiv:2501.00049)	January 2025	Encoder–decoder Seq2Seq with LSTM cells and attention, GloVe embeddings, Keras/TensorFlow; trained on 3,700 tourism Q&A utterances.	94.12% test accuracy; no dependence on paid third-party NLP APIs (lower cost, no vendor lock-in); tailored to domain and local context.	Small single-domain dataset with a 98/1/1 split (tiny test set); overfitting in some configurations (72.43% test accuracy in one); single-turn answers, multi-turn dialogue left as future work.
6	Apprentice Bot Model Design and Implementation for Psychological Clients' Therapy (PsychoBot)	Gidey, K., Beyene, H., Haileslassie, F. & Berihu, H. (2025)	Discover Applied Sciences, 7(3), 170	2025 (Vol. 7, Issue 3)	Hybrid generative + retrieval chatbot using Seq2Seq, Amharic Word2Vec embeddings and memory networks; compares LSTM, GRU, double GRU and transposed GRU.	Targets a low-resource language; systematic comparison of RNN variants; transposed GRU performed best (79.86% accuracy, validation loss 0.208).	Moderate accuracy (about 80%); long training time (1,434 min for the best model); no user-friendly web interface reported.
7	Conversational Chatbot for Enhancing Healthcare Services	Kaur, A. & Prakash, G. (2026)	Springer, Decision Sciences for Quality and Productivity Improvement, pp. 3–26	July 2026	End-to-end healthcare chatbot with multi-attention (transformer-style) mechanism for intent classification, slot extraction, dialogue management and response generation.	Captures different aspects of an utterance and long-range dependencies that earlier recurrent models overlook; covers the full dialogue pipeline.	Not RNN-based, so included as the attention/transformer contrast; heavier compute than RNN models; quantitative results not reviewed here.
8	An Intelligent Chatbot Design and Implementation Model Using Long Short-Term Memory with Recurrent Neural Networks and Attention Mechanism	Choudhary, P. & Chauhan, S. (2023)	Decision Analytics Journal, 9, 100359	2023	LSTM Seq2Seq with attention, Bag of Words and beam-search decoding; trained on a 3,725 Q&A dialog dataset; evaluated with BLEU-N.	Reported 59.44% BLEU-N improvement over baseline chatbots; beam search improves response quality.	Focuses on linguistic precision; no user-experience study or real-world deployment context.
9	A Generative Model Based Chatbot Using Recurrent Neural Networks	Raj, V. & Phridviraj, M. S. B. (2023)	Springer, CCIS Vol. 1798 (ANTIC conference)	2023	Open-domain generative chatbot using deep LSTM Seq2Seq with attention, trained on Reddit conversations; evaluated with a Turing test against Cleverbot.	Can answer questions no rule was written for; generative rather than template-based.	Open-domain generation can be inconsistent or irrelevant; Turing-test evaluation is subjective; compared against a single baseline.
10	Designing Effective Chatbot System Using GRU with Beam Search	Thorat, S. A., Jadhav, V. D. & Mane, D. T. (2022)	ICTACT Journal on Soft Computing, 13(1), 2750–2755	October 2022	GRU encoder–decoder with attention and beam-search decoding; trained on the Cornell Movie Dialogue Corpus.	GRU is lighter than LSTM; reported better accuracy and lower loss than LSTM Seq2Seq models.	Movie-dialogue corpus is not task-specific; evaluation limited to accuracy and loss, with no user study.

Note: Refs 2 and 6 are summarized from abstracts and secondary descriptions, and Refs 7–10 partly from abstracts. Check details against the full papers before submission.

Research Gap

Synthesizing the limitations reported across the ten reviewed studies, the following gaps emerge in current RNN-based chatbot research:

- **Long-context and multi-turn handling remains weak.** Simple RNNs struggle with extended dialogues (Ref 4), repetition appears in generated text (Ref 1), and multi-turn capability is explicitly left as future work (Ref 5).
- **RNNs are rarely used alone.** Recent systems pair RNNs with GPT models (Ref 4), transformers or reinforcement learning (Ref 2), or are replaced by attention-based models (Ref 7), so the standalone strength of RNN chatbots is not clearly established.
- **Small, narrow, single-domain datasets.** Training data ranges from 109 FAQ pairs (Ref 4) to about 3,700 utterances (Ref 5), with a 1% test split in Ref 5, limiting generalization and statistical confidence.
- **Retrieval-based vs. generative trade-off unresolved.** Retrieval-based RNN chatbots give reliable but fixed answers (Ref 3), while generative ones can be flexible but inconsistent (Refs 1, 9).
- **Limited real-world and user-centered validation.** Several studies report only accuracy, loss or BLEU (Refs 5, 8, 10), use subjective Turing-style testing (Ref 9), or test with very few users (Ref 4: 32 participants).
- **No unified evaluation standard.** Accuracy, BLEU-N, F1, Likert satisfaction and Turing tests are used in different studies, making cross-paper comparison difficult.
- **Low-resource and multilingual support is underexplored.** Only Ref 6 addresses a low-resource language (Amharic), and its accuracy is still moderate (about 80%).
- **Safety and reliability in sensitive domains.** Health and mental-health chatbots (Refs 2, 3, 4, 6) need clinical-grade validation and ethical safeguards, yet the studies mostly report accuracy or satisfaction rather than outcomes; data bias and ethics are also flagged in Ref 1.
- **Dependence on external APIs and heavy infrastructure.** Hybrid designs rely on paid third-party models (Ref 4), while lightweight, self-contained RNN chatbots that avoid this dependency (Ref 5) are validated only on small datasets.
- **Accuracy vs. efficiency trade-off.** Lightweight RNNs train quickly and run on modest hardware (Ref 4), but attention and transformer models capture context better at higher computational cost (Ref 7).

SDG Mapping

The reviewed literature aligns with the following United Nations Sustainable Development Goals (SDGs):

SDG	Relevant Studies	Mapping Rationale
SDG 3 – Good Health and Well-Being	Refs 2, 3, 4, 6, 7	Mental-health support, health-insurance guidance and patient-conversation chatbots improve access to health information and support.
SDG 4 – Quality Education	Refs 1, 10	Chatbots that assist research writing, summarization and academic feedback support learning and research skills.
SDG 8 – Decent Work and Economic Growth	Refs 4, 5, 8, 9	Automating customer service, FAQ handling and tourism information raises productivity and supports local economic activity.
SDG 9 – Industry, Innovation and Infrastructure	Refs 1–10 (all)	RNN, LSTM, GRU, Seq2Seq and attention models are innovations in AI-driven digital service infrastructure.
SDG 10 – Reduced Inequalities	Refs 3, 6	Voice-enabled and local-language (Amharic) chatbots widen access for users with limited literacy or language resources.
SDG 16 – Peace, Justice and Strong Institutions	Ref 4	Chatbots that explain public health-insurance coverage make institutional information more transparent and accessible.

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